

Chapter 50 — Renal Dysgenesis and Cystic Disease of the Kidney

- **Congenital and acquired renal malformations form a wide spectrum.** This chapter covers disorders producing **dysgenesis, cystic lesions, or both.**
- **Many arise from maldevelopment of the kidney and urinary tract** through anomalous genetic mechanisms, some understood and some not. Molecular genetics underpins normal development, agenesis, and the effect of mutant signalling proteins on maldevelopment and cyst formation.

OVERVIEW OF GENETICS

- **Urinary tract anomalies** (kidney, ureter, bladder, urethra) arise from **genetic and environmental factors together.**
- **Genetic factors = chromosomal abnormalities plus single-gene mutations.** Complex gene–gene and gene–environment interactions also operate.

SINGLE-GENE MUTATIONS AND THEIR RENAL PHENOTYPES

Gene	Phenotype
PAX2	Renal hypodysplasia or aplasia
EYA1	Branchio-oto-renal (BOR) syndrome
HNF1B	Renal cysts and diabetes syndrome
PKD1, PKD2 (primary cilium)	Polycystic kidney disease
Susceptibility variants	Urinary tract infection (UTI) → renal scarring and long-term damage

RENAL AGENESIS AND DYSPLASIA

- **Terminology is used loosely and confusingly.** This chapter uses the **American Academy of Pediatrics (AAP) Section on Urology definitions (1987).**

Renal Agenesis

- Covered in Chapter 49.

Renal Dysplasia

Definition

- **Renal dysgenesis = maldevelopment affecting kidney size, shape or structure.** Its **two primary subtypes are dysplasia and hypoplasia.**
- **Dysplasia ("abnormal tissue") is strictly a histological diagnosis:** embryonic, immature mesenchyme and primitive renal components. Metanephric differentiation halts partway, usually from an error in the **wolffian duct–ureteric bud (UB)–metanephric blastema interaction.**
- **Histology:**
 - distorted architecture;
 - immature or primitive glomeruli;
 - nephron precursors (**comma- and S-shaped bodies**);
 - **cartilage;**

- **primitive ducts:** tubules encircled by collars of fibromuscular cells (Fig. 50.1). **Ericsson and Ivemark (1958): the primitive duct is the sine qua non of all dysplasia.**
- **Cysts may or may not be present.**
- **Cause is twofold:**
 1. a **primary, inherent (often genetic) defect** in nephron and collecting-duct differentiation;
 2. a **secondary defect from congenital urinary tract obstruction.**

Aetiology

- **Renal dysplasia is the primary cause of childhood end-stage renal disease (ESRD).**
- **Two theories:**

DYSPLASIA — PRIMARY VS SECONDARY THEORY

Primary UB failure	Fetal outflow obstruction
Intrinsic failure of ureteric bud activity	Obstruction disrupts renal development

- **Associated syndromes:** Fraser, BOR, renal-coloboma, Kallmann, Simpson-Golabi-Behmel, Smith-Lemli-Opitz, HDR (hypoparathyroidism, sensorineural deafness, renal dysplasia) and Townes-Brocks.
 - The kidney may be mostly normal with dysplastic areas, entirely dysplastic, or absent.
- **Ultrasound appearance (Fig. 50.3):**
 - **small kidney, hyperechoic relative to the liver;**
 - **loss of corticomedullary differentiation;**
 - little distinction from perirenal fat;
 - $\pm$ cysts.
- **Not all cysts are dysplastic, and not all dysplasia is cystic.**

KEY POINTS: DYSPLASIA

- **Hallmark: the primitive duct**, a duct encircled by a collar of fibromuscular cells.
- **Dysplasia is a histopathological, not a clinical, definition.**
- **Often associated with UB anomalies and/or obstruction.**

HYPOPLASIA AND HYPODYSPLASIA

Renal Hypoplasia

- **Hypoplasia = underdevelopment from a deficient number of cells. Renal hypoplasia = reduced functioning renal mass.**
- **Hypodysplastic kidneys usually accompany ectopic ureteric orifices. The degree of dysplasia correlates with the degree of ectopia.** A few have normal orifices, with or without obstruction.
- **Assessment:**
 - **ultrasound measurement** (the size is reduced in all forms; calyces may be dilated);
 - function biochemically;
 - **radionuclide study if unilateral.**

Oligomeganephronia

- **First described in 1962. Hypoplasia with a reduced number of nephrons, each hypertrophied.**

- **Differs from simple hypoplasia:** in simple hypoplasia renal mass is reduced but nephron number is normal.
- **Usually bilateral.** Unilateral cases are reported with contralateral agenesis.
- **Course:**
 - **after the first year:** short stature, polyuria and polydipsia, proteinuria;
 - function subnormal but stable for years;
 - **creatinine clearance falls rapidly in the teens**, with marked proteinuria (about 2 g/24 h) and progressive tubular loss;
 - **progressive, with many reaching renal failure in the second decade.**

Evaluation

- **Proteinuria is the first laboratory sign, preceding functional decline by years.**
- Then raised urea and creatinine, **hyponatraemia, metabolic acidosis.**
- **Small kidneys on renal ultrasound (RUS) usually establish hypoplasia.**

Treatment

- **Supportive:** maintain biochemical balance, haemoglobin and growth.
 1. **high fluid intake;**
 2. **correct salt loss and acidosis;**
 3. **limit protein to 1.5 g/kg/day** in the stable phase;
 4. **angiotensin-converting enzyme (ACE) inhibitor even if normotensive** to slow progression.
- **ESRD → dialysis and transplantation.** A living related donor is acceptable because the disease is not familial.

Ask-Upmark Kidney (Segmental Hypoplasia)

- **Segmental renal hypoplasia:** an extreme form of **acquired reflux nephropathy**. Patients are **usually young females with hypertension.**
- **Mostly acquired:** chronic atrophic pyelonephritis from vesicoureteric reflux (VUR). Some are congenital; segmental arterial anomalies have also been proposed.

Clinical Features

- **Usually aged ≥10 years at diagnosis.**
- **Severe hypertension ± headache, hypertensive encephalopathy; retinopathy in about half.**
- **Usually unilateral.** Bilateral disease brings proteinuria and renal insufficiency.
- **Classic radiological sign: deep, narrow segmental "slit-scars" in the midzone.** They are visible on ultrasound, computed tomography (CT) or magnetic resonance imaging (MRI) but often overlooked.

Histopathology

- **Small kidney (12–35 g). Deep grooves on the lateral convexity.**
- **Parenchyma beneath the groove: thyroid-like tubules**, sharply demarcated from normal kidney (Fig. 50.5: absent glomeruli, thick-walled arteries).
- Thin medullary band; remnants of the corticomedullary junction and arcuate arteries.
- **Arteriosclerosis common; juxtaglomerular hyperplasia** may be seen.

Treatment

- **Nephrectomy normalises blood pressure (BP) regardless of plasma renin activity.**

- **Unilateral disease** → **partial or total nephrectomy**. If BP fails to respond, suspect an **unrecognised contralateral scar or generalised arteriosclerosis**.
- **Bilateral disease with insufficiency** → **medical management**, with dialysis or transplantation if needed.
- **Correcting reflux may prevent further renal damage but will not cure the hypertension.**

Renal Hypodysplasia

- **Accompanies:** primary obstructive megaureter, **ureterocele**, **posterior urethral valves (PUV)** and **prune-belly syndrome**.

HYPODYSPLASIA AND THE POSITION OF THE URETERIC ORIFICE

Lateral ureteral ectopia	Medial ureteral ectopia (± ureterocele)
VUR → reflux-associated hypodysplasia	Hypodysplasia (± obstruction)

- **Cortex:** thin from hydronephrosis, or severely dysplastic with many small cysts.
- **PUV — two types of hypodysplasia** (Osathanondh and Potter; eFig. 50.2):

HYPODYSPLASIA IN POSTERIOR URETHRAL VALVES — TWO FORMS

	Milder form	Severe form
Cysts	Small, usually subcapsular	Larger, widespread, with islands of cartilage
Function	Near normal	Impaired
Obstruction and reflux	Less severe	Earlier onset, more severe

- **Prune-belly syndrome** (absent abdominal musculature; triad syndrome):
 - grossly deformed kidneys with variable hypodysplasia;
 - **wide, tortuous ureters with large, laterally placed orifices;**
 - **usually no lower-tract obstruction;** urethral obstruction or atresia only in the most severe cases.

FORMS OF RENAL DYSGENESIS COMPARED

Entity	Defining feature	Laterality	Hallmark	Course
Dysplasia	Histological: primitive ducts, cartilage	Uni- or bilateral	Primitive duct	Leading cause of childhood ESRD
Simple hypoplasia	Reduced mass, normal nephron number	—	Small kidney	Depends on extent
Oligomeganephronia	Fewer, hypertrophied nephrons	Usually bilateral	Proteinuria first	ESRD in the second decade
Ask-Upmark	Segmental hypoplasia (reflux nephropathy)	Usually unilateral	Slit-scar , thyroid-like tubules	Severe hypertension; nephrectomy cures
Hypodysplasia	Hypoplasia plus dysplasia	Follows the anomaly	Ectopic orifice, obstruction	Degree ∝ degree of ectopia

KEY POINTS: HYPOPLASIA AND HYPODYSPLASIA

- **Hypoplasia and hypodysplasia are associated with ectopic orifices** and sometimes obstruction.
- **Oligomeganephronia:** fewer nephrons, each hypertrophied.

- **Ask-Upmark kidneys** show segmental hypoplasia, usually with reflux, and often cause **severe hypertension**.

Concept — In dysgenesis, the ureteric orifice predicts the kidney. The bud that emerges from the wrong place on the wolffian duct meets the wrong blastema, so *ectopia*, *obstruction* and *dysplasia* travel together. The further the *ectopia*, the worse the kidney.

CYSTIC DISEASES OF THE KIDNEY

- **The kidney is one of the commonest sites of cyst formation.** Renal cystic diseases are sporadic or genetic, congenital or acquired, united by cysts in one or both kidneys, and **often need a multidisciplinary approach**.
- **Renal cysts** derive mainly from tubules:
 - a layer of partially de-differentiated epithelium around urine-like fluid or semisolid material;
 - they can arise **anywhere from Bowman capsule to the papillary tip**;
 - they may or may not communicate with the nephron, or may lose an earlier connection.
- **Exceptions to tubular origin:**
 - **multicystic dysplasia arises before nephrons form** (abnormal metanephric induction, primary blastemal defect, or early obstruction);
 - **benign multilocular cyst is a neoplasm**.
- **Three processes build and enlarge a cyst** (Fig. 50.6):
 1. **epithelial proliferation** in a tubule segment, starting in a single cell after a "**second hit**" disables the normal allele;
 2. **fluid accumulation** in the expanding segment;
 3. **disturbed extracellular matrix** (thickened basement membrane, interstitial inflammation and fibrosis).
 - The cyst then **detaches from its tubule**, and **net epithelial secretion fills it**.
- **Fluid mechanism:** beyond the loop of Henle, tubular cells can secrete solute and fluid. This normally loses to the stronger **sodium (Na⁺) absorption**. **When Na⁺ absorption falls, net NaCl and fluid secretion occurs**.
- **The primary cilium is central:** structural or functional ciliary defects drive cyst formation and abnormal epithelial behaviour. Newly identified signalling pathways offer **targeted therapy** to slow cystogenesis.

Classification

- **Primary division: genetic (inheritable) vs non-genetic (non-heritable)** (Box 50.1).

MULTICYSTIC VS POLYCYSTIC

Multicystic	Polycystic
Dysplastic kidney from aberrant development	Kidney developed normally, no dysplasia , nephrons throughout
—	"Polycystic kidney disease" = ARPKD and ADPKD ; many progress to renal failure as nephrons become diseased

- **Tuberous sclerosis and von Hippel-Lindau (VHL) disease: hyperplastic cysts with normal nephrons.** Renal failure occurs only when cysts or tumours compress nephrons.
- **Non-heritable neoplasms:** benign multilocular cyst, cystic renal cell carcinoma (RCC).
- **Medullary sponge kidney:** a disease mainly of **dilated ectatic collecting ducts**, with initially normal nephrons.

BOX 50.1 CYSTIC DISEASES OF THE KIDNEY

Inheritable	Non-heritable
Autosomal recessive (infantile) polycystic kidney disease	Multicystic kidney (multicystic dysplastic kidney)
Autosomal dominant (adult) polycystic kidney disease	Benign multilocular cyst (cystic nephroma)
Juvenile nephronophthisis (autosomal recessive) and medullary cystic disease (autosomal dominant) complex	Simple cysts
Congenital nephrosis (familial nephrotic syndrome) (autosomal recessive)	Medullary sponge kidney
Familial hypoplastic glomerulocystic disease (autosomal dominant)	Sporadic glomerulocystic kidney disease
Multiple malformation syndromes with renal cysts (e.g., tuberous sclerosis, VHL disease)	Acquired renal cystic disease; calyceal diverticulum (pyelogenic cyst)

Inheritable Cystic Disease

- **Classified by mode of transmission:** autosomal dominant, autosomal recessive, X-linked and others. Causes are single-gene, X-linked or chromosomal (Table 50.1).

TABLE 50.1 CHARACTERISTICS OF MAJOR INHERITABLE AND NONHERITABLE CYSTIC KIDNEY DISEASES

Disease	Chromosome	Renal findings	Extrarenal manifestations
Inheritable			
ARPKD	6	Newborn: large, homogeneous, echogenic kidneys	Congenital hepatic fibrosis ; biliary dysgenesis
ADPKD	PKD1: 16; PKD2: 4; PKD3 not mapped	Cysts throughout parenchyma; large kidneys	Diverticulitis; liver, spleen, pancreatic cysts; mitral valve prolapse; berry aneurysms
Juvenile nephronophthisis (AR)	2	Corticomedullary cysts developing after onset of renal failure; tubular basement membrane always thickened	Retinitis pigmentosa (16%; Senior-Løken) ; rarely skeletal, hepatic fibrosis, Bardet-Biedl, ocular motor apraxia, neurological
Medullary cystic disease (AD)	Not mapped	Corticomedullary cysts developing before renal failure ; basement membrane may not be thickened	None
Tuberous sclerosis (AD)	TSC1: 9; TSC2: 16	Cysts and angiomyolipomas throughout; cysts even in utero; RCC 3% (table) vs 2% (text)	Adenoma sebaceum, epilepsy, intellectual disability, cranial tumours
VHL disease (AD)	3	Cysts, adenomas, clear cell RCC 35–38% (table) vs about 50% (text)	Cerebellar haemangioblastoma, retinal angioma, phaeochromocytoma , pancreatic and epididymal cysts
Non-heritable			
MCDK	—	Diffuse cysts, early metanephric remnants, little or no nephron development; commonest renal cystic disease in newborns	Unusual
Benign multilocular cyst	—	Benign cystic neoplasm ; rest of the kidney normal; boys <4 years, women >30 years	None
Simple cysts	—	Single or multiple; nephrons normal; common with age	None

Disease	Chromosome	Renal findings	Extrarenal manifestations
Medullary sponge kidney	—	Ectatic collecting ducts; nephrons usually normal	None
Acquired renal cystic disease	—	Diffuse cysts, adenomas, occasionally RCC; increases with duration of ESRD	None

- **Use:** the **extrarenal column discriminates** — liver fibrosis → ARPKD; berry aneurysm → ADPKD; retinitis pigmentosa → NPH; angiofibroma → TSC; haemangioblastoma → VHL.
- **Traps:** **chromosome 16 carries PKD1 and TSC2. RCC figures differ between table and text** (TSC 3% vs 2%; VHL 35–38% vs about 50%), probably older series and a different denominator versus lifetime incidence.

AUTOSOMAL RECESSIVE (INFANTILE) POLYCYSTIC KIDNEY DISEASE

- **Autosomal recessive polycystic kidney disease (ARPKD):** rapid, symmetrical, bilateral renal enlargement in infancy from **collecting-duct cysts**. It is **invariably accompanied by congenital hepatic fibrosis**.
- **Gene:** PKHD1 on chromosome 6.
- **"Infantile" is a misnomer:** it spans a severity spectrum. **The earlier it appears, the more severe it is;** if not apparent at birth it appears later in childhood (to about 13 years, rarely 20).
- **Incidence 1 in 10,000–50,000 live births** (Table 50.2: 1 in 5000–40,000). **Up to half of affected newborns die in the first days**, so incidence among 1-year survivors is lower.
- **Of neonatal survivors, about half are alive at 10 years.**

Clinical Features

- **In utero:** enlarged echogenic kidneys and **oligohydramnios** → **Potter facies**, limb deformity, **pulmonary hypoplasia** and respiratory distress.
- **Newborn:** **enormous, kidney-shaped, non-bosselated, hard flank masses that do not transilluminate**, sometimes large enough to obstruct delivery. Creatinine and urea rise soon after birth.
- **30–50% die shortly after birth** of uraemia or respiratory failure.
- **All have congenital hepatic fibrosis**, with variable biliary ectasia and periportal fibrosis.
- **Survivors of the neonatal period** have milder disease, often reaching adulthood. Some reach 3–4 years before renal failure.
 - **Hypertension and renal insufficiency dominate;** liver disease grows with age;
 - chronic renal insufficiency brings growth failure, anaemia and osteodystrophy;
 - **renal calcification is common.**
- **Later-onset disease:** slower renal failure and hypertension; **problems are hepatic** — **portal hypertension, oesophageal varices, hepatosplenomegaly. Hepatocellular function and liver enzymes stay normal.** Kidneys may look normal.
- **No association with renal neoplasms.**

Evaluation

- **Suspected on prenatal ultrasound**, often with oligohydramnios.
- **Fetal and newborn ultrasound:** **bilateral, very large, diffusely echogenic kidneys** (brighter than liver).
 - The echogenicity comes from **innumerable microcysts (compacted, dilated collecting ducts) creating countless interfaces;**
 - **pyramids blend into the cortex** → a **homogeneous kidney** (Fig. 50.8A).

- **ADPKD in the newborn usually shows diffuse, larger cysts**, but can look identical (Fig. 50.8B).
- **Differential diagnosis:** severe bilateral hydronephrosis (hypoechoic calyces); **multicystic kidney** (cysts in a non-reniform mass); sporadic glomerulocystic kidney disease (GCKD); bilateral mesoblastic nephroma; **Wilms tumour**; bilateral renal vein thrombosis.
- **CT if in doubt:** it is more sensitive to inhomogeneity, and so to tumour.

Treatment

- **No cure.**
 - **respiratory support**;
 - treat hypertension, heart failure, and renal and hepatic failure;
 - **portal hypertension** → decompression (splenorenal shunt);
 - **varices** → **endoscopic sclerotherapy** (or gastric transection and reanastomosis);
 - **haemodialysis and transplantation** eventually.
- **Nephrectomy** — unilateral or bilateral, **for respiratory or nutritional compromise, BP control, or to make room for peritoneal dialysate**. The evidence is limited, and the risks are real:
 - loss of nephron mass and earlier need for kidney replacement;
 - **arterial hypotension after bilateral nephrectomy, with neurological injury** in infants on peritoneal dialysis.
- **ARegPKD registry:**
 - **very early bilateral nephrectomy (VEBNE, within 3 months)** carried the **most severe neurological complications**;
 - **VEBNE and hypotensive episodes were independent risk factors**;
 - early bilateral nephrectomy and very early dialysis also fared worse than controls with large total kidney volume.
- **Verdict: be very cautious with bilateral nephrectomy in the first months.** The remaining kidney may grow rapidly after unilateral nephrectomy and force a second operation, so **buy time where possible**.

KEY POINTS: AUTOSOMAL RECESSIVE POLYCYSTIC KIDNEY DISEASE

- **PKHD1, chromosome 6; all have congenital hepatic fibrosis.**
- **Severe form in utero or infancy;** milder later in childhood, rarely to 20 years.
- **Early disease: very large, homogeneously hyperechoic kidneys (microcysts);** discrete cysts appear with age; overlap with ADPKD.
- **Bilateral nephrectomy only with caution.**

Concept — ARPKD is a disease of the collecting duct and the bile duct together, so kidney and liver share its whole course. It kills neonates through pulmonary hypoplasia (oligohydramnios) and uraemia, and older survivors through the liver (portal hypertension). Severity is fixed by how early it declares itself.

AUTOSOMAL DOMINANT (ADULT) POLYCYSTIC KIDNEY DISEASE

- **Autosomal dominant polycystic kidney disease (ADPKD) is by far the commonest inheritable renal cystic disease: about 1 in 400–1000 live births** (Table 50.2: 1 in 500–1100).
- **Causes 7–15% of patients on haemodialysis.**

- **Penetrance theoretically 100%; half of offspring affected; about 10% sporadic (no family history). 96% manifest clinically by 90 years.**
- **Usually identified in the fourth–fifth decade**, but it occurs in neonates and infants. **Everyone affected manifests it if they live long enough**, but **renal failure is rare before 40 unless infantile onset, which is far more aggressive.**
- **Associated anomalies:** cysts of liver, pancreas, spleen and lungs; **berry aneurysms of the circle of Willis;** colonic diverticula; aortic aneurysms; **mitral valve prolapse.**

TABLE 50.2 COMPARISON OF AUTOSOMAL RECESSIVE AND AUTOSOMAL DOMINANT POLYCYSTIC KIDNEY DISEASE

	ARPKD	ADPKD
Gene defect	Chromosome 6	Chromosomes 16 and 4
Incidence (table)	1:5000–40,000	1:500–1100
Incidence (body text)	1:10,000–50,000	1:400–1000
Usual age at presentation	Perinatal	Third to fifth decades
Ultrasound	Symmetrically enlarged, homogeneous, hyperechoic	Large cystic kidneys, sometimes asymmetrical
Histology	Collecting-duct ectasia; cysts from the collecting duct	Micro- and macrocysts from the entire nephron
Liver	Always congenital hepatic fibrosis (variable severity)	Cysts, mostly adults
Other organs	None	Berry aneurysms, colonic diverticula, mitral valve prolapse, cysts (seminal vesicle, arachnoid, pancreas)

- **Pattern:** the **segment of origin** (collecting duct vs whole nephron) explains the ultrasound (homogeneous microcysts vs discrete macrocysts); the liver lesion differs in kind (fibrosis vs cysts).
- **Traps:** incidence differs between table and text (different series; both shown). **Neonatal ADPKD can mimic ARPKD exactly → scan the parents.**

Genetics

PKD1 VS PKD2

	PKD1	PKD2
Locus	Short arm of chromosome 16	Long arm of chromosome 4
Protein	Polycystin-1	Polycystin-2
Share of ADPKD	About 85%	About 15%
Cysts	Usually by age 20 : most children by 10, nearly all by 20	Later
ESRD	Fifth decade (50s)	Seventh decade (70s)

- **PKD3** accounts for a very small number with neither mutation.
- **PKD1 and PKD2 act in the same pathway**, hence the similar phenotype. **Polycystins inhibit cell proliferation through several pathways.**
- **Contiguous gene syndrome:** PKD1 lies immediately beside TSC2 on chromosome 16. A large deletion removes both, producing **tuberous sclerosis plus the ADPKD renal phenotype.**

Clinical Features

- **Highly variable expression**, from fetal death to normal function into old age. **Intrafamilial variability** implicates genetic background and environment.
- **Family ultrasound screening now finds asymptomatic children with cysts.**
- **Usually bilateral**, but **about 17% asynchronous or asymmetrical** ("unilateral ADPKD"), especially in children. The other kidney becomes involved later.
- **Symptoms typically at 30–50 years**: microscopic or gross haematuria, flank pain, gastrointestinal symptoms (renomegaly or diverticula), colic (clot or stone), hypertension.
- **Haematuria in about 50%**; the **presenting sign in about a fifth to a third**.
- **Anaemia is unusual even in ESRD**: the increased renal mass raises **erythropoietin**.
- **Pain (flank or abdominal) is the commonest presenting symptom in adults**. Causes: mass effect, cyst haemorrhage, infection (including infected cysts), stones.
- **Stones in 20–30%**. **Uric acid and calcium oxalate stones equally common**.
 - Treat **conservatively where possible**: alkalinisation, spontaneous passage, **extracorporeal shock wave lithotripsy (ESWL)**.
 - **Hydronephrosis is a less useful stone sign** because cysts camouflage it.
- **UTI commoner in women**. **Suspected cyst infection → consider cyst aspiration**, which is diagnostic and therapeutic.
- **Hypertension is now a common presentation**. Figures:
 - **about 50% at 20–35 years with normal renal function**;
 - **more than 60% before renal impairment overall**;
 - **nearly 100% with ESRD**.
 - Mechanism: **renin-mediated, from stretching of intrarenal vessels around cysts → distal ischaemia**.
 - **Control is critical** to slow renal failure and reduce cardiac and aneurysmal morbidity.
- **Renal failure**: function is usually preserved despite accelerating cyst growth **until the fourth–sixth decade**.
 - **Risk factors for earlier ESRD**: **PKD1, male sex, first haematuria <30 years, hypertension onset <35 years, hyperlipidaemia, sickle cell trait**.
 - **Function declines in proportion to kidney and cyst volume**. Hypertension and vascular remodelling add to the decline.

Extrarenal Manifestations

- **Hepatic cysts: the commonest extrarenal manifestation**.
 - rare in children, and rising with age to **almost all by 50** (up to 90% of adults);
 - in PKD1 and non-PKD1 genotypes;
 - **more prevalent and larger in females (oestrogen effect)**;
 - lined by biliary-type epithelium, with fluid like the bile-salt-independent fraction of bile;
 - usually asymptomatic (mass, infection, haemorrhage).
 - **Rarely → portal hypertension and variceal bleeding**, which can then blur the ADPKD distinction.
 - **Portal hypertension is far commoner in ADPKD** (always from congenital hepatic fibrosis). Congenital hepatic fibrosis very rarely accompanies ADPKD, mainly with perinatal diagnosis.
- **Intracranial aneurysms (ICA)**, mainly **berry aneurysms of the circle of Willis**:
 - **in 10–30%**; **about 9% of these die of subarachnoid haemorrhage (SAH)**;

- **SAH presents as a sudden severe headache ± nausea and vomiting → immediate evaluation;**
- **risk rises with family history of ruptured ICA and with PKD1;**
- MRI detects small aneurysms (average about 6 mm at diagnosis);
- **small (<1 cm) aneurysms rupture less often, unless there is a family history of rupture or ADPKD itself;**
- **BP control is critical.** Some bleeds are hypertensive intracerebral haemorrhages rather than aneurysmal.
- **Other cysts: seminal vesicles (40%)** — rarely infertility but sperm motility may suffer; **arachnoid (8%); pancreas (5%)** — both asymptomatic.
- **Mitral valve prolapse and colonic diverticulosis.** Diverticulosis marks patients more likely to have hepatic cysts and symptomatic berry aneurysms.

Association With Renal Cell Carcinoma

- **Renal adenomas are almost as common as in acquired renal cystic disease (ARCD): about 1 in 4–5 patients.**
- **But RCC incidence in ADPKD is no higher than in the general population.** Contrast ESRD with ARCD, where RCC runs at 3–6× the general population.
- **When RCC occurs in ADPKD it differs:**

RCC IN ADPKD VS THE GENERAL POPULATION

Feature	ADPKD	General population
Age	Younger	Older
Bilateral synchronous	12%	1–5%
Multicentric	28%	6%
Sarcomatoid	33%	1–5%

- **Diagnosis is hard:** tumour hides among complex cysts, haemorrhage, clot, debris and infection.

Evaluation

- **Take at least a three-generation family history** (renal disease, hypertension, strokes). **Counsel before any imaging or testing.**
- **Genetic testing:** when **imaging is equivocal** and a definite diagnosis is needed.

GENETIC TESTING IN ADPKD — BENEFITS VS HARMS

Benefits	Harms
Informs family planning	Discrimination (insurance, employment)
Early detection and treatment of complications	Psychological stress of anticipated organ failure
Selects unaffected relatives as living related donors	—

- **Ultrasound (Ravine) criteria** for a person at 50% risk (Fig. 50.11):

RAVINE ULTRASOUND CRITERIA FOR ADPKD (AT-RISK INDIVIDUALS)

Age	Criterion
<30 years	≥2 cysts, unilateral or bilateral
30–59 years	≥2 cysts in each kidney

Age	Criterion
≥60 years	≥4 cysts in each kidney

- **Sensitivity:** nearly 100% at ≥30 years and in younger PKD1 carriers, but only about two-thirds in PKD2 carriers <30. A negative scan in a young person does not exclude PKD2.
- **No family history → presumptive diagnosis** needs **bilateral renal cysts plus ≥2 of:**
 - bilateral renal enlargement;
 - ≥3 hepatic cysts;
 - cerebral artery aneurysm;
 - solitary cyst of the arachnoid, pineal, pancreas or spleen.
- **CT or MRI** is better than ultrasound for **extrarenal cysts** (Fig. 50.12).
 - **CT diagnoses cyst haemorrhage:** acute blood 50–90 Hounsfield units (HU), higher than old blood;
 - **MRI needs no contrast** — useful in impaired function. **Haemorrhagic cysts are bright on T1 and dark on T2.**
- **Infantile presentation:** half have visible macrocysts. **The rest can be indistinguishable from ARPKD → examine the parents.**

Treatment and Prognosis

- **No cure; aim to limit complications and delay ESRD.**
- **Hypertension:**
 - more than 60% hypertensive before renal impairment;
 - it worsens renal function, causes cardiac disease, and predisposes to intracranial haemorrhage. **Control it rigorously.**
 - **Agent: no proven drug of choice, but ACE inhibitors or angiotensin receptor blockers (ARB) are logical first line.** The hypertension is renin-mediated; they raise renal blood flow, have few side effects, and may be renoprotective beyond BP.
 - **Standard target <130/80 mm Hg;** the optimal target is unsettled (see HALT-PKD).
- **Chronic pain:**
 - **exclude and treat infection, stone and tumour;**
 - **avoid nephrotoxic analgesics (NSAIDs);**
 - **limit opioids to acute episodes** (dependence risk).
- **When conservative measures fail:**
 1. **ultrasound- or CT-guided cyst aspiration** (diagnostic and therapeutic);
 2. **surgical unroofing** of multiple or large cysts — laparoscopic, or open via flank or dorsal lumbotomy. It **improves pain only**, neither accelerating nor preserving renal decline, and has variable BP effect;
 3. **nephrectomy for symptomatic patients with ESRD.**
- **Upper-tract infection** (common, especially in women):
 - **no response to appropriate antibiotics → suspect a non-communicating infected cyst;**
 - **lipophilic antibiotics penetrate cysts:** trimethoprim–sulfamethoxazole, chloramphenicol, fluoroquinolones;
 - **failure → percutaneous or surgical drainage;**
 - **exclude obstruction, abscess and stones.**
- **Berry aneurysm screening:**

SCREENING FOR INTRACRANIAL ANEURYSM IN ADPKD

Screen (CT or MR angiography)	Do not screen
Family history of aneurysm or SAH	Low-risk patients (aneurysms rare; those found rarely rupture)
Previous aneurysm rupture	—
Before major elective surgery	—
High-risk occupation (e.g., airline pilot)	—
High anxiety about ICA	—

- **Aneurysm management:**

- **<7 mm, found on screening → conservative;**
- **>7–10 mm (rupture up to 2% per year) or symptomatic → open clipping or endovascular coiling;**
- small asymptomatic aneurysms are treated only after a previous bleed.
- **Rupture tracks size and poorly controlled BP.**

Emerging Therapeutics

- **Aim: alter cyst growth.** Targets are the renin–angiotensin system and **cyclic adenosine monophosphate (cAMP).**

DISEASE-MODIFYING TRIALS IN ADPKD

Trial	Agent and comparator	Endpoint	Result
HALT-PKD (largest PKD trial)	Lisinopril + telmisartan vs lisinopril alone	Total kidney volume (TKV)	Combination did not alter TKV growth in early ADPKD
HALT-PKD (BP arm)	Rigorous BP (95–110/60–75) vs standard (120–130/70–80 mm Hg)	TKV, estimated glomerular filtration rate (eGFR), left ventricular mass index (LVMI), albuminuria	Slower TKV growth; no change in eGFR; greater fall in LVMI and albuminuria
TEMPO, REPRISE	Tolvaptan (selective vasopressin V2-receptor antagonist; lowers renal cAMP) vs placebo	TKV, renal function	Slowed TKV growth and functional decline

- **Tolvaptan's costs:**

- **aquaresis** (thirst, polyuria, nocturia, polydipsia from electrolyte-free water loss);
- **liver function abnormalities;**
- slightly more serious events (chest pain, headache);
- **high discontinuation rate.**

- **Food and Drug Administration (FDA) approval April 2018: tolvaptan is the first US treatment for adults with ADPKD.**

KEY POINTS: AUTOSOMAL DOMINANT POLYCYSTIC KIDNEY DISEASE

- **Commonest inheritable cystic disease; usually apparent after 30, but any age including in utero.**
- **99% PKD1 (chromosome 16) or PKD2 (chromosome 4);** PC1 and PC2 inhibit proliferation.
- **Cysts on ultrasound before 20 in almost all; symptoms at 30–50.**
- **Liver cysts in virtually all by 50; berry aneurysms 10–30% (about 9% of these die of SAH).**

- **RCC not increased. Treat with tight BP control, ACE inhibitor and/or ARB, possibly tolvaptan.**

Concept — ADPKD is a volume disease. Cyst expansion stretches vessels (renin-mediated hypertension), compresses nephrons (function falls in proportion to kidney volume) and hides tumours and stones. Every current therapy — BP control, ACE inhibition, tolvaptan — is judged by whether it slows the growth of total kidney volume.

JUVENILE NEPHRONOPHTHISIS AND MEDULLARY CYSTIC DISEASE COMPLEX

- Juvenile nephronophthisis (NPH; Fanconi, 1951) and medullary cystic kidney disease (MCKD; Smith and Graham, 1945) are inheritable ciliopathies.
- **Shared hallmark: interstitial fibrosis.** They differ in inheritance, age at ESRD and extrarenal involvement.
- **Shared features:**
 - polyuria and polydipsia in more than 80% (less than in diabetes insipidus), from a vasopressin-resistant concentrating defect with salt wasting → a large dietary salt intake is often needed;
 - renal failure 5–10 years after presentation;
 - pathology identical:
 - grossly, small-to-normal kidneys with corticomedullary cysts;
 - histologically, a triad: (1) irregular thickening and disintegration of the tubular basement membrane; (2) tubular atrophy with cysts; (3) interstitial infiltration and fibrosis.

JUVENILE NEPHRONOPHTHISIS VS MEDULLARY CYSTIC KIDNEY DISEASE

	Juvenile NPH	MCKD
Inheritance	Autosomal recessive	Autosomal dominant
Frequency	Commoner: 10–20% of childhood renal failure; 1–5% of dialysis or transplant patients	Less common
ESRD	Mean 13 years, almost always before 25	Milder; manifest in all by 50; ESRD in the third–fourth decade
Hypertension	Absent	Can be significant
Other features	—	Type 1: slowly progressive chronic kidney disease (CKD), minimal proteinuria and haematuria, later hypertension and hyperuricaemia. Type 2: hyperuricaemia characteristic; gout in teenage males
Extrarenal	In 20%: retinitis pigmentosa 16% (renal–retinal = Senior–Løken syndrome); skeletal, hepatic fibrosis, Bardet–Biedl, ocular motor apraxia (Cogan), Joubert-spectrum neurological defects, cone-shaped epiphyses, ataxia	Kidney only

- **Bardet–Biedl syndrome** = obesity, intellectual disability, polydactyly, retinitis pigmentosa, hypogenitalism.
- **Diagnosis is difficult early** (vague symptoms) → high index of suspicion.
 - Ultrasound often misses the small cysts; it may show small kidneys (NPH) and hyperechogenic parenchyma from tubulointerstitial fibrosis (Fig. 50.13).
 - Contrast CT or MRI is more sensitive for corticomedullary and medullary cysts, but absent cysts do not exclude the diagnosis.

- **Treatment of NPH is supportive:**
 - **sodium replacement early;**
 - **avoid unnecessary salt restriction and diuretics** (volume contraction, azotaemia);
 - then **dialysis and transplantation.**
 - **Allografts do not recur:** there are no anti-basement-membrane antibodies.
 - **Prostaglandin E2 receptor agonists** are a potential therapy for NPH-associated ciliopathies.

KEY POINTS: JUVENILE NEPHRONOPHTHISIS AND MEDULLARY CYSTIC DISEASE COMPLEX

- **Juvenile NPH (autosomal recessive):** 10–20% of childhood renal failure; **ESRD in the early teens.**
- **MCKD (autosomal dominant):** manifests in early adulthood.
- **Polyuria and polydipsia are the hallmarks of both; hypertension occurs in MCKD, not NPH.**
- **Near-identical histology:** severe interstitial nephritis, corticomedullary cysts.

Concept — NPH and MCKD are one lesion (tubulointerstitial fibrosis with a salt-wasting concentrating defect) in two inheritances. Recessive disease is early and syndromic; dominant disease is later and kidney-only. Because they waste salt, these are the cystic kidneys in which salt restriction and diuretics harm.

MULTIPLE MALFORMATION SYNDROMES WITH RENAL CYSTS

- **Renal cysts feature in several multiple-malformation syndromes.**
 - The **autosomal dominant tuberous sclerosis and VHL disease** are those urologists meet;
 - **autosomal recessive** examples: **Meckel syndrome, Jeune asphyxiating thoracic dystrophy, Zellweger cerebrohepatorenal syndrome;**
 - many involve **glomerular cysts**, some cystic dysplasia.

Tuberous Sclerosis Complex

- **Usually autosomal dominant neurocutaneous disorder; up to 1 in 6000.** It is characterised by hamartomas in nearly every organ.
- **Genes: TSC1 (chromosome 9) and TSC2 (chromosome 16).**
- **Classic triad:**
 - **epilepsy (about 80%);**
 - **intellectual disability (about 60%);**
 - **adenoma sebaceum (facial angiofibromas, about 75%):** firm red or brown telangiectatic papules of the **nasolabial folds, chin and cheeks**, appearing **after 2 years** and persisting.
 - Autism and neurobehavioural disability are also associated.
- **Earliest skin sign: the white "ash-leaf" macule.** **Ultraviolet light examination** reveals lesions earlier and belongs in the work-up.
- **Diagnosis now rests on major and minor features, not the triad. Tuberous sclerosis complex (TSC) requires 2 major, or 1 major + 2 minor:**

TUBEROUS SCLEROSIS COMPLEX — DIAGNOSTIC FEATURES

Major features	Minor features
Renal angiomyolipoma (AML)	Multiple renal cysts
Facial angiofibromas or forehead plaque	Non-renal hamartoma

Major features	Minor features
Non-traumatic ungual or periungual fibroma	Hamartomatous rectal polyps
≥3 hypomelanotic macules	Retinal achromic patch
Shagreen patch	Cerebral white-matter radial migration tracts
Multiple retinal nodular hamartomas	Bone cysts
Cortical tuber; subependymal nodule; subependymal giant cell astrocytoma	Gingival fibromas
Cardiac rhabdomyoma	"Confetti" skin lesions
Lymphangiomyomatosis	Multiple enamel pits

- Kidneys may be normal, or contain cysts, AMLs or both (Figs 50.14 and 50.15).

Clinical Features

- Renal involvement is second only to the central nervous system: AML, cysts and RCC.

ANGIOMYOLIPOMA — TUBEROUS SCLEROSIS VS SPORADIC

	AML in TSC	Sporadic AML
Frequency	Very common (40–80%)	Uncommon
Number	Bilateral, multiple	Usually single
Sex	Both sexes	Mainly middle-aged women
Age	Rare before 6, common after 10 years	Middle age

- The main risk is haemorrhage or mass effect; also mass, tenderness, hypertension, renal insufficiency. AMLs alone probably do not cause renal failure.
- Pleomorphism and mitoses look aggressive, but AMLs never metastasise.

Association With Renal Cell Carcinoma

- More than coincidental: about 2% (Table 50.1: 3%). This is well below VHL and ARCD.
- Compared with sporadic RCC: female predominance, younger age, more bilateral.
- Suspect RCC in an enlarging lesion without demonstrable fat, or with calcification. Early detection is essential.

Radiographic Evaluation

- **Ultrasound:** cysts sonolucent; AML fluffy and white (echogenic).
- **Cysts without AML can mimic ADPKD** → apparent ADPKD may declare itself as TSC years later.
 - CT helps: AML in kidney or other organs → TSC; cysts in other organs → ADPKD.
 - AML has negative attenuation (fat, about –50 HU); a cyst is near water (Fig. 50.15).
 - Head CT or MRI shows calcified tubers or gliosis.
- **Fat-poor AML** — harder to diagnose, but **benign; no extirpative surgery needed**. Subtypes:
 1. hyperattenuating;
 2. isoattenuating;
 3. AML with epithelial cysts. - They have characteristic perfusion features on dedicated CT. Missing classic or fat-poor AML leads to unnecessary surgery.

Treatment

- More patients now survive their neurological disease, so urologists manage the kidneys.

- **Renal disease was the leading cause of death in one series:** metastatic RCC, AML haemorrhage, and (most often) renal failure from cysts and AMLs.
- **AMLs are benign and usually need no treatment. Large AMLs bleed.**
- **Older advice** (see table) also included **annual RUS or CT, renal-sparing surgery for pain, haemorrhage, function-threatening growth or possible RCC, and warning about pregnancy and oestrogen** with multiple or large AMLs.
- **International TSC Consensus Group — at diagnosis:**
 - **abdominal imaging at any age: MRI preferred** — 25–30% of AMLs are fat-poor and missed on ultrasound; CT next best;
 - **intravenous contrast** helps identify cysts and fat-poor AML, and is likely less harmful than feared when **glomerular filtration rate (GFR) ≥ 60 mL/min/1.73 m²**;
 - **MRI also finds** aortic aneurysms, hepatic hamartomas and pancreatic neuroendocrine tumours;
 - **accurate BP** (secondary hypertension risk);
 - **GFR** by creatinine or cystatin C equations.

ANGIOMYOLIPOMA IN TSC — OLDER ADVICE VS CONSENSUS GUIDELINE

	Older advice	International TSC Consensus
Threshold	>4 cm	>3 cm and growing, asymptomatic
First line	Prophylactic embolisation or surgical excision	mammalian target of rapamycin (mTOR) inhibitor (everolimus), with safe-use protocols including pre-immunisation
Second line	—	Selective embolisation then corticosteroids, kidney-sparing resection, ablation of exophytic lesions (if mTOR inhibitor contraindicated)
Acute haemorrhage	Renal-sparing surgery	Embolisation then corticosteroids
Nephrectomy	Bilateral before transplant considered	Avoid: complications, later insufficiency, poor CKD prognosis

- **Why mTOR inhibition wins:** continued tolerability is preferable to the renal damage of AML growth or of embolisation or ablation. **Everolimus has remained safe and effective** for TSC or sporadic lymphangioleiomyomatosis AML on long-term follow-up.
- **Surveillance:**
 - **kidney function, proteinuria and BP at least annually** (more often if abnormal); age-specific BP criteria in children;
 - **imaging every 1–3 years:**
 - **yearly** if a tumour is **approaching 3 cm or growing**;
 - **longer intervals** if **no tumours, or only small (<1 cm) stable ones**;
 - look for polycystic change, RCC and AML growth.
- **Hypertension:** renin–angiotensin–aldosterone system inhibitor first line. In the young, **ACE inhibitors and ARBs are associated with less AML formation.**
- **Advanced renal failure:** mTOR inhibitors still help prevent bleeding and may slow decline. **Offer all renal replacement options, including to patients with neurological impairment.**

von Hippel–Lindau Disease

- **Autosomal dominant:**
 - **cerebellar and retinal haemangioblastomas;**

- **cysts of pancreas, kidney and epididymis; epididymal cystadenoma;**
- **phaeochromocytoma;**
- **clear cell RCC.**
- **All ethnic groups, both sexes; 1 in 30,000–50,000; >95% penetrance by 65 years.**
- **Grouped with the phakomatoses** (Sturge–Weber, TSC, neurofibromatosis): shared inheritance pattern and neoplastic propensity.

Aetiology

- **Germline mutation of the VHL tumour-suppressor gene, chromosome 3 (3p25).**
- **pVHL forms a complex that marks hypoxia-inducible factors (HIF) for degradation.**
 - **HIF-1 α** induces angiogenic factors — **vascular endothelial growth factor (VEGF)**, most prominent in RCC; **platelet-derived growth factor (PDGF)**; **transforming growth factor- β (TGF- β).**
 - **Unsuppressed, it drives haemangioblastoma and RCC.**

HIF REGULATION — NORMAL CELL VS VHL-DEFICIENT CELL

	Normal cell, normoxia	Hypoxia, or VHL-deficient cell (even in normoxia)
pVHL	Functional	Absent or reduced (hypoxia) / defective
HIF	Destroyed via VHL	Stabilised, active
Consequence	Angiogenic genes silent	VEGF, PDGF, TGF-β up; defective extracellular matrix assembly

- **pVHL is also a ciliary protein controlling ciliogenesis.** Its loss impairs cilia in renal cancer cells; re-expression rescues them.

Clinical Features

- **Mean presentation 35–40 years; no sex preference** for the disease or its RCC. Contrast sporadic RCC: male, usually after 50.

VHL DISEASE — MANIFESTATIONS

Manifestation	Frequency and features
Renal cysts	Commonest and often earliest (76%); bilateral 75%, multifocal 87%; typically large; rarely an ADPKD picture or renal failure
RCC	About 50% (text; Table 50.1: 35–38%); usually asymptomatic; haematuria if rupturing into the collecting system. Metastatic RCC: 30–40% of deaths (key points)
Phaeochromocytoma	10–17% (text; key points 10–12%); clusters in specific families
Cerebellar haemangioblastoma	Most frequent, characteristic and specific tumour; symptomatic at 15–40 years; benign but more than half of deaths
Retinal angioma (haemangioblastoma)	Early; bleeding → blurred vision, detachment, blindness
Pancreatic cysts	Up to 70%, usually benign; islet cell tumours about 12%; metastatic RCC to pancreas not unusual
Endolymphatic sac tumour	Papillary tumour of the petrous bone → hearing loss
Epididymal papillary cystadenoma	10–26% of males, uni- or bilateral; no malignant potential → examination and ultrasound; surgery only for severe pain. Female counterpart: broad-ligament cystadenoma

- **Urologist's role: surveillance to catch small RCCs before metastasis. Metastatic RCC competes with haemangioblastoma as the leading cause of death → annual or twice-yearly CT.**

Evaluation

DIAGNOSIS OF VHL DISEASE — WITH VS WITHOUT FAMILY HISTORY

Known family history	No family history
One retinal or cerebellar haemangioblastoma, RCC or pheochromocytoma	≥2 haemangioblastomas (retinal or cerebellar), or 1 haemangioblastoma plus one other characteristic lesion

- **Ultrasound:** benign cysts show no internal echoes, well-defined margins and acoustic enhancement.
- **CT:** sharp thin walls, homogeneous contents, **no enhancement**. **CT is often more useful than ultrasound for multiple lesions, and images the adrenals** (pheochromocytoma).
- **Small lesions:** tumour and cyst are indistinguishable → regular **thin-collimation CT**. **Larger or suspected RCC → magnified or subtraction renal angiography.**
- **MRI is poor for small renal tumours** (signal like parenchyma on T1 and T2) unless renal contour is altered.

Screening

- **Molecular testing now restricts screening to genetically affected relatives.** Previously all at-risk relatives needed fundoscopy and serial CT.

TABLES 50.4 AND 50.5 SCREENING PROTOCOL FOR PATIENTS AND FAMILIES WITH VON HIPPEL–LINDAU DISEASE (MAHER ET AL., 1990) VS SURVEILLANCE GUIDELINES TO HELP PREVENT SEVERE VHL COMPLICATIONS (VHL ALLIANCE, 2020)

Modality (target)	Table 50.4 — affected persons	Table 50.4 — relatives at risk	Table 50.5 — VHL Alliance 2020 (by starting age)
History and examination	Annual	Annual	Yearly from age 1
BP and pulse (pheochromocytoma or paraganglioma)	—	—	Yearly from age 2
Eye (retinal haemangioblastoma)	Annual fundoscopy	Annual, 10–60 years	Dilated eye examination every 6–12 months from before age 1; yearly from 30
Metanephrines (pheochromocytoma or paraganglioma)	Annual 24-hour urine	Annual	Yearly from 5 years
Brain (± spine) MRI (CNS haemangioblastoma)	Every 3 years from 10–50, then every 5 years (gadolinium MRI or CT)	Every 3 years from 15–40, then every 5 years to 60	MRI brain and spine every 2 years from 11
Audiogram (endolymphatic sac tumour)	—	—	Every 2 years from 11
MRI internal auditory canal	—	—	Once, at 15
Abdomen (RCC, pheochromocytoma or paraganglioma, pancreatic tumours or cysts)	Annual RUS + CT every 3 years (more often if multiple renal cysts)	Annual RUS + CT every 3 years, 20–65	MRI abdomen every 2 years from 15
From 65 years	—	—	Stop routine metanephrines and MRI; continue yearly examination, BP and eye checks

Modality (target)	Table 50.4 — affected persons	Table 50.4 — relatives at risk	Table 50.5 — VHL Alliance 2020 (by starting age)
Pregnancy	—	—	Everything before conception (not audiogram); eye examination every 6–12 months during pregnancy

- **Shift:** from CT and ultrasound starting at 10–20 years to MRI from infancy, organ by organ, stopping at 65, with a pre-conception check.
- **Pattern:** eye first (retinal angiomas earliest), then brain and ear from 11, abdomen from 15.
- **Traps:** the old protocol screens every at-risk relative, the modern approach **only genetic carriers**; **abdominal imaging is now MRI every 2 years**, whereas the text's annual or twice-yearly CT reflects the older protocol.

Treatment

- **Multidisciplinary.** Tumours are **multiple, bilateral and recurrent**.
- **Mainstay:** close surveillance and minimising surgery.
- **Goal in VHL (and all hereditary cancer syndromes):** cancer control, not cure, while preserving renal and adrenal parenchyma.
- **Kidney:**
 - **resection accepts that microscopic disease is left behind;**
 - **open nephron-sparing surgery is the standard of care for low-grade RCC in VHL.** It is **not curative:** it spares parenchyma and prevents metastasis of existing lesions;
 - **local recurrence 75–85%, yet 10-year disease-specific survival is 81–94%;**
 - **radical (bilateral) nephrectomy** classically gave only about 50% survival and brings dialysis, transplantation and poor quality of life. It is **reserved for high-grade disease;**
 - **laparoscopic and image-guided ablation** (radiofrequency, stereotactic radiotherapy, cryoablation) is **still under investigation.**
- **Adrenal:**
 - **bilateral pheochromocytomas are frequent;** bilateral adrenalectomy leaves lifelong steroid replacement and poorer quality of life;
 - so **preserving adrenocortical function is a primary goal** → **laparoscopic partial adrenalectomy** for small, non-functioning tumours. **Recurrence about 15%.**

KEY POINTS: MULTIPLE MALFORMATION SYNDROMES

- **TSC (autosomal dominant; TSC1 chromosome 9, TSC2 chromosome 16):** epilepsy, intellectual disability, adenoma sebaceum; **renal cysts about 20%, AML 40–80%, RCC about 2%.**
- **VHL (autosomal dominant; chromosome 3):** haemangioblastomas; pancreatic, renal (76%) and epididymal cysts; pheochromocytoma; **clear cell RCC about 50% (30–40% of deaths).**
- **pVHL regulates HIF targets;** its loss raises growth factors, **VEGF foremost in RCC.**

Concept — Both syndromes grow tumours from normal nephrons, so the kidney fails only when lesions compress or replace it. The urologist's task in both is *preservation over extirpation*: medical shrinkage of AML in TSC, and nephron-sparing control rather than cure of RCC in VHL.

MULTICYSTIC DYSPLASTIC KIDNEY

- **Multicystic dysplastic kidney (MCDK):** a developmental anomaly of **multiple cysts of varying size with no identifiable normal parenchyma**.
 - **No function by definition.** The contralateral kidney is usually normal and **compensatorily hypertrophied**.
 - **Only unilateral disease is compatible with life.**
- **Morphology:**
 - usually non-reniform, with an **atretic ureter** and **no glomerulus–calyx connection**;
 - **"bunch of grapes"** with little stroma between cysts (Fig. 50.17);
 - size from a nubbin to a mass filling the abdomen.

MCDK — THE TWO TYPES

Infundibulopelvic type	Hydronephrotic type
Commoner; atresia of renal pelvis and ureter	Less common; only the upper ureter atretic

- **Both types:** ureters absent or atretic; **renal vessels hypoplastic**.
- **Rarely:** after complete obstruction by a **ureterocele**; in **one half of a duplex** (usually the upper pole with an **ectopic ureter**); or in a **horseshoe kidney**.

Aetiology

- **Obstruction theory:** an extreme form of obstructive hydronephrosis from pelvic or ureteric atresia.
- **UB theory (supported):** mutations in **EYA1, SIX1, WNT, WT1, GDNF, AT2 and PAX2** — UB development genes found in syndromic dysplasia. This also **explains the frequent contralateral anomalies**.

Clinical Features

- **Commonest renal cystic disease; second commonest abdominal mass in infants.**
- **Most diagnosed prenatally; 1 per 1000–4000 live births.** Male preponderance (55–70%); side preference disputed.
- **Contralateral tract frequently abnormal:** pelviureteric junction (PUJ) obstruction 3–12%; VUR 18–43%.
- **Natural history benign:**
 - complications extremely rare;
 - **about 40% involute spontaneously;**
 - **large series show no increased risk of hypertension or neoplasm** (hypertension is recognised but rare).
- **Follow-up conservatively:** the primary care provider monitors **BP, abdominal mass and UTI**.

Evaluation

- **Prenatal detection → repeat postnatal ultrasound** in the first few days and at 1 month. This confirms the diagnosis and assesses the contralateral kidney and bladder.

ULTRASOUND: MCDK VS SEVERE PUJ OBSTRUCTION (FIG. 50.19)

	MCDK	PUJ obstruction (hydronephrosis)
Cyst distribution	Haphazard , various sizes	Peripheral calyces organised around a central pelvis
Central or medial large cyst	Absent	Present (the pelvis)
Communication between cysts	None visible	Demonstrable

	MCDK	PUJ obstruction (hydronephrosis)
Small cysts among large ones	Present	Absent
Isotope uptake	Seldom any (Fig. 50.20)	Usually some function

- **Radioisotope scan** (dimercaptosuccinic acid, DMSA, or technetium-99m mercaptoacetyltriglycine, ^{99m}Tc-MAG3) resolves difficult cases.
- **Rarely needed:** angiography (small or absent artery); cystoscopy (hemitrigone and absent orifice, or an orifice with atresia on retrograde study).
- **Voiding cystourethrogram (VCUG):**
 - often done because of **reflux into the solitary functioning kidney**;
 - of little value if serial high-quality ultrasound shows a normal bladder and contralateral kidney in a child without UTI.

Treatment and Prognosis

- **"Ignore the multicystic kidney unless its bulk is inconvenient";** attend to the **contralateral tract**.
- **Nephrectomy if:** mass effect on respiration or feeding; the rare associated hypertension; spontaneous or traumatic cyst rupture (pain, haemorrhage); **increasing solid tissue**.
- **Mistaking a non-functioning hydronephrotic kidney for MCDK has little consequence.** Totally non-functioning PUJ kidneys are rarely salvageable, and complete non-function is itself unusual in PUJ obstruction.
- **Malignant transformation: extremely low, perhaps nil.**
- **Ultrasound surveillance every 3–12 months is common practice, but there is no evidence it helps or is cost-effective.**

KEY POINTS: MULTICYSTIC DYSPLASTIC KIDNEY

- **Multiple cysts, no normal parenchyma;** active nephrogenesis-gene expression; **changing morphology**.
- **Usually shrinks or disappears (then imaged as renal aplasia);** very occasionally grows; **very rarely associated with Wilms tumour**.
- **No clear indication for removal unless solid tissue increases.**
- **No increased risk of hypertension or neoplasm** in large series.

Concept — *MCDK is a non-functioning kidney that removes itself. Management therefore looks away from it, to the solitary contralateral kidney and its reflux or obstruction. It is removed only for mechanical complications or new solid tissue.*

BENIGN MULTILOCLAR CYST (CYSTIC NEPHROMA)

- **Multilocular cystic neoplasm of the kidney.** It sits on a **spectrum** of four lesions (Fig. 50.21):
 1. **benign multilocular cyst** (most benign);
 2. **multilocular cyst with partially differentiated Wilms tumour**;
 3. **multilocular cyst with nodules of Wilms tumour**;
 4. **cystic Wilms tumour** (most malignant).
- **Not a segment of multicystic kidney:** the two differ clinically, histologically and radiologically.

- **Nature disputed:** segmental dysplasia, hamartoma or neoplasm. The confusion comes from **variable histology** (primitive stroma, tubular and even muscle maturity, epithelial atypia) between and within lesions.

MULTILOCULAR CYSTIC LESIONS — PAEDIATRIC VS ADULT SPECTRUM (FIG. 50.21)

Behaviour	Paediatric	Adult
Benign	Benign multilocular cyst	Benign multilocular cyst
Acts benign	Cystic partially differentiated Wilms tumour	Multilocular cystic RCC
Malignant (nodules present)	Multilocular cyst with nodules of Wilms tumour	Multilocular cyst with nodules of RCC (existence uncertain)
Malignant	Cystic Wilms tumour	Cystic RCC

- **No lesion converts into another. Tumour nodules = malignant**, but the prognosis of nodular and cystic forms is **better than that of the solid equivalents**.

Clinical Features

- **95% present before 4 years or after 30 years.**

BENIGN MULTILOCULAR CYST — CHILD VS ADULT

	Child (<4 years)	Adult (>30 years)
Sex	Male 2:1	Female 8:1
Presentation	Asymptomatic flank mass	Flank mass, abdominal pain, haematuria (cyst herniating through urothelium into the pelvis)
Surgery	Nephrectomy usual	Partial nephrectomy more often feasible (more normal tissue)

Evaluation

- **Tests:** ultrasound, CT, MRI, cyst puncture and aspiration, double-contrast cystography.
- **Ultrasound and CT distinguish multilocular cyst from multicystic kidney, but cannot separate** multilocular cyst from its Wilms or adenocarcinoma variants, mesoblastic nephroma, cystic Wilms tumour or **clear cell sarcoma**.
- **Ultrasound: highly echogenic septa, sonolucent loculi** (debris can look solid).
- **CT:** septa less dense than parenchyma. **Calcification rarely visible in children.**

Treatment

- **Nephrectomy for any multilocular cystic lesion, even the most benign.** Partial nephrectomy or excision if localised with well-preserved tissue.
- **Nodules of Wilms tumour or cystic Wilms tumour → treat as Wilms tumour per National Wilms Tumor Study (NWTs) staging**, recognising the favourable outlook.
- **Cystic RCC → treat as malignant**, recognising its better prognosis.

KEY POINTS: BENIGN MULTILOCULAR CYST

- **Benign, non-dysplastic neoplasm: boys <4 years, women >30 years.**
- **Diagnosis only by surgical excision.**

- **Benign end of a spectrum** ending at cystic Wilms tumour.

Concept — *Imaging cannot tell the benign multilocular cyst from its malignant relatives, so the lesion is treated by what it might be. Excision is both diagnosis and treatment, and the nodules in the specimen decide whether it is managed as cancer.*

SIMPLE CYSTS

- **Commonest renal cystic lesion.** Oval or round, solitary or multiple, uni- or bilateral, filled with **clear or straw-coloured plasma-like fluid**. **Not connected to the nephron** (though perhaps nephron-derived).
- **Age-related:**

SIMPLE CYSTS — INCIDENCE BY AGE

Age	Incidence
In utero	Seen from 14 weeks
Birth–18 years	Stable, 0.1–0.45% (mean 0.22%)
By 40 years	About 20%
After 60 years	Up to 50%

Clinical Features

- **Usually incidental** on ultrasound, CT or urography.
- **Possible:** mass or pain; **haematuria from rupture** into the collecting system; **hypertension from segmental ischaemia**; **calyceal or pelvic obstruction**. May or may not enlarge.

Evaluation

SIMPLE CYST CRITERIA — ULTRASOUND VS CT

Ultrasound (all four required)	CT
No internal echoes	Homogeneous contents, –10 to +20 HU (water)
Sharply defined, thin, distinct, smooth wall	Sharp, thin, smooth wall and margin
Good through-transmission with posterior acoustic enhancement	No enhancement after IV contrast
Spherical or slightly ovoid	Spherical or ovoid
All met → malignancy negligible (Fig. 50.23)	All met → accuracy approaches 100% (Fig. 50.24: 17 HU before, 16 HU after contrast = no enhancement)

- **Criteria not met** (septa, irregular margin, calcification, suspicious areas) → **CT, or needle aspiration or MRI**.
- **A cluster of cysts also needs study:** it may hide a small carcinoma, and **CT beats ultrasound** here.
- **Enhancement means vascular tissue** (or contrast mixing with fluid). **Enhancing wall nodules or thick septa = vascularity** → **high suspicion of neoplasm**.
- **Hyperdense cysts (20–90 HU)** are still simple if all of these hold:
 - **no enhancement**, and other CT and ultrasound criteria met;
 - **homogeneous on narrow window settings**;
 - **≤3 cm**;
 - **≥one-quarter of the circumference projecting beyond the renal contour** (so the wall can be assessed).

- **Otherwise consider complicated cysts (blood, pus, calcium) or cystic neoplasm.**
- **Cyst puncture is now rarely needed.** Indications:
 1. **suspected infection** (diagnostic and therapeutic);
 2. **low-level echoes on ultrasound but a classic cyst on CT;**
 3. **borderline lesion in a poor surgical candidate.**
- **Calcification: 2% of simple cysts vs 10% of RCCs.** It is **peripheral** in cysts and **central** in tumours.
- **Numerous bilateral cysts without liver cysts** can mimic ADPKD. **Do not diagnose ADPKD** in doubtful cases without documented autosomal dominant family history or genetic confirmation.
- **Children:**
 - **stable (no growth for 2 years) simple cysts need no further follow-up;** many resolve;
 - **no malignancy in simple cysts, and a low rate in complex ones;**
 - **simple cyst <15 mm, asymptomatic, age <18 → repeat ultrasound at 2 years and discharge if stable;**
 - **continue surveillance for cysts ≥15 mm, complex cysts, family history of inheritable cystic disease, and concomitant genitourinary anomalies.**

Classification

- **Bosniak classification:** proposed 1986, clarified 1997, **category IIF added by Israel and Bosniak 2003** (Box 50.2).

BOX 50.2 BOSNIAK'S CLASSIFICATION OF SIMPLE AND COMPLEX CYSTS

Category	Features	Management
I	Simple benign cyst: good through-transmission, no internal echoes, sharp smooth wall	No surgery
II	Minimally complicated: fine septa, minimal (fine or short-segment) calcification, high density, infection	No surgery (explore only if septa numerous, irregular or thick)
IIF	Cannot be classed II or III: more, even thick or nodular, calcification; septa minimally enhancing, calcified septa non-enhancing	No surgery, but follow — malignant progression in about 11% (Fig. 50.25: diffuse calcification, non-enhancing septa)
III	Indeterminate: more calcification, thicker or irregular wall, prominent septa (multilocular-like), chronically infected thick wall. Features also seen in malignancy; more likely benign than malignant	Surgical exploration and/or removal, individualised (Gerota exposure or partial nephrectomy)
IV	Clearly malignant: large cystic components, irregular margins, solid enhancing vascular elements	Surgical removal (radical nephrectomy)

- **Pattern:** the **decisive feature is enhancement, not calcification;** increasing calcification is of little concern, **thickening or irregularity of wall or septa is.**
- **Traps:** IIF = **follow-up, not surgery,** yet about 1 in 10 progress; **III is more often benign but is managed surgically.**
- **Paediatric modifications:**
 - **classes I–II mostly benign → periodic ultrasound if unchanged and asymptomatic;**
 - **III–IV warrant excision (>90% intermediate or malignant pathology).** The modified system gives reasonable risk stratification.

- **Bosniak version 2019:**

- **Goals:** (1) curb **over-diagnosis and over-treatment of cystic RCC**; (2) reduce **inter-reader variability**; (3) sharpen **malignancy rates per class**; (4) **raise specificity** to spare benign masses treatment (morbidity, lost function, cost).
- **Changes:** (1) **formally incorporates MRI**; (2) **defines imaging features and classes explicitly**; (3) covers more masses seen in practice (e.g., incompletely characterised, highly likely benign cysts); (4) **moves more masses into lower classes**.

Treatment and Prognosis

- **Once malignancy is excluded, an asymptomatic cyst needs no surgery.**
- **Symptoms:**
 - pain (exclude other causes such as stones) — from haemorrhage, or calyceal or infundibular obstruction;
 - rare hypertension;
 - **infection (rare, severe: fever, flank pain, sympathetic pleural effusion).**
- **Treat the symptom:**
 1. **percutaneous aspiration ± sclerosant** (especially after re-accumulation). Sclerosants: glucose, phenol, iophendylate, bismuth phosphate, absolute ethanol — **none dominant**;
 2. **surgical or laparoscopic unroofing** (trans- or retroperitoneal);
 3. percutaneous resection or intrarenal marsupialisation;
 4. **infected cyst → drainage plus antibiotics.**

KEY POINTS: SIMPLE CYSTS

- **Seldom in children; increasingly common with age in adults.**
- **Benign simple cyst on ultrasound:** no internal echoes; sharp, thin, smooth wall; spherical or oval; good transmission with acoustic enhancement. **Otherwise → contrast CT.**
- **Bosniak CT classification guides exploration or removal of complex cysts.**

Concept — *A cyst is benign because it has no blood supply, so every imaging criterion is a test for vascularity. Enhancement, nodularity and wall thickening signal living tissue. Bosniak simply grades how much living tissue the cyst might contain.*

MEDULLARY SPONGE KIDNEY

- **Medullary sponge kidney (MSK; precalyceal canalicular ectasia):** dilated distal collecting ducts with cysts and diverticula confined to the medullary pyramids. Usually **non-heritable**.
- **Appearance:** the ducts look like the "**bristles on a brush**"; when more ectatic and calcified, like a "**bouquet of flowers**".

Clinical Features

- **Usually benign, possibly lifelong-silent. Presents after 20 years.**
- **Presentations:** renal colic 50–60%, UTI 20–33%, gross haematuria 10–18%.
- **MSK in stone formers:** 2.6–21%, higher in women. **UTI also commoner in women.**
- **Metabolic:** hypercalciuria in one-third to one-half; incomplete distal renal tubular acidosis (RTA) in 30–40% (→ hypocitraturia).

Diagnosis

- **Urographic triad:**
 1. **enlarged kidneys** ± papillary calcification;
 2. **elongated papillary tubules or cavities filling with contrast;**
 3. **papillary blush and persistent medullary opacification.**
- **Intratubular calcium** appears as stones or **nephrocalcinosis**.

Treatment and Prognosis

- **Manage stones and infection.**

MEDULLARY SPONGE KIDNEY — AGENTS FOR STONE PREVENTION

Agent	Target	Indication	Caution
Potassium citrate	Hypocitraturia (incomplete RTA), hypercalciuria	First choice: reduces calciuria and stone recurrence	—
Thiazide	Hypercalciuria	Lowers calciuria and stone formation. Give to stone formers even without hypercalciuria	—
Inorganic phosphate	Hypercalciuria	If thiazide cannot be used	Not with urease-producing infection (struvite risk)
Fluids and low-sodium diet	Urine volume, calciuria	All	—

KEY POINTS: MEDULLARY SPONGE KIDNEY

- **Usually non-heritable;** dilated collecting ducts look like **brush bristles on intravenous pyelogram**, sometimes calcified.
- **Renal colic 50–60%, UTI 20–33%, gross haematuria 10–18%, hypercalciuria about 33%.**

SPORADIC GLOMERULOCYSTIC KIDNEY DISEASE

- **"Glomerulocystic" = diffuse, bilateral cysts of the glomeruli or Bowman space.** The term is often used loosely for any condition with glomerular cysts, which are **common to many cystic diseases**.
- **Sporadic GCKD proper: non-heritable; bilaterally enlarged kidneys with small cysts, predominantly of Bowman space.**
- **Often indistinguishable from other glomerular-cyst conditions; only the absence of affected relatives and anomalies separates it:**

GLOMERULAR-CYST DIFFERENTIAL DIAGNOSIS

Condition	Inheritance	Discriminator
Sporadic GCKD	None	No family history, no anomalies
ADPKD	Autosomal dominant	Family history; extrarenal cysts, aneurysms
Familial hypoplastic glomerulocystic disease	Autosomal dominant	Family history; small kidneys
Juvenile NPH	Autosomal recessive	Salt wasting, ESRD in teens, retinitis pigmentosa

ACQUIRED RENAL CYSTIC DISEASE

- **Acquired renal cystic disease (ARCD; acquired cystic kidney disease, ACKD):** bilateral cystic change in ESRD not due to inherited cystic disease.
- **Diagnosis:** ≥ 3 cysts in each kidney.
- **Described in 1977 in haemodialysis;** almost as common on peritoneal dialysis.
- **Dialysis is not a prerequisite: uraemia causes it.** Dialysis merely prolongs survival and gives the cysts time to form.

ARCD PREVALENCE BY DURATION OF RENAL FAILURE

Stage	Prevalence
ESRD before dialysis	About 10%
3 years on dialysis	44%
5 years	60%
≥ 10 years	>90%

- **Male 3:1;** generally unrelated to age or cause of renal failure.
- **Children on dialysis: similar high incidence (about 20–45%),** rising with dialysis duration.
- **Effect of transplantation — the source conflicts:**
 - **regression after transplantation** is reported by several groups (key points: "cysts usually regress");
 - **more recent data do not support delay or reversal;**
 - **new cysts form in 18% after transplantation.**
 - **What is agreed: malignant potential persists.**

Aetiology

- **Tubular obstruction** (fibrosis, oxalate crystals, vascular occlusion, ischaemia) → cysts.
- **Uraemic toxin:** cysts regress after a good transplant and **return when a failed graft forces a return to dialysis.** The graft appears to clear a cystogenic or carcinogenic uraemic toxin.
- **Growth factors:** epidermal growth factor (EGF) and other uraemic stimulants.
- **Renotrophic agents** released after loss of renal tissue → hyperplasia, cysts and, in extreme cases, tumours.

Clinical Features

- **Mostly asymptomatic.**
- **Commonest presentation:** loin pain and/or haematuria from spontaneous bleeding into cysts (up to 50%). It may cause a **subcapsular or retroperitoneal haematoma** from a cyst or RCC, **worsened by uraemic coagulopathy and dialysis heparin.**
- **Less common:** cyst infection, stones, a **rising haematocrit** (erythropoietin).
- **Image anyone with microscopic haematuria (>5 red cells per high-power field), and dialysis and transplant patients** (tumour risk).
- **Neoplasms:**
 - **adenomas in about 10% on chronic haemodialysis, 20–25% with ARCD;**
 - **overall RCC prevalence on haemodialysis about 1%.**

RENAL MALIGNANCY RISK IN END-STAGE RENAL DISEASE — THE FIGURES AND THEIR DENOMINATORS

Comparison	Relative risk
RCC with ARCD vs without (dialysis patients)	3×
RCC in large vs small cystic kidneys	6×
ESRD with ARCD vs general population (ADPKD section)	3–6×
Renal malignancy in all dialysis patients vs general population	5–50×
RCC after renal transplantation vs general population	>15×

- The ranges differ because the comparison groups differ (ARCD only, all dialysis, post-transplant). **They are not contradictory estimates of one quantity.**

Evaluation

- **Ultrasound:** commonest for diagnosis and monitoring. It shows **small hyperechoic kidneys with cysts** of various sizes.
- **CT and MRI find more cysts; MRI is best for small lesions.**
- **CT with and without contrast** is best for **telling a few simple cysts from ARCD** and **best for detecting neoplasms.**
- **Pre-dialysis ESRD:**
 - **avoid iodinated contrast** (residual function);
 - use ultrasound or gadolinium MRI instead, **with caution: gadolinium in ESRD risks nephrogenic systemic fibrosis (NSF)**, rare but life-threatening.
- **Uraemic patient with fever → consider an infected cyst.** Internal echoes or a thick wall on ultrasound suggest it; CT shows wall thickening; **cyst puncture confirms and identifies the organism.**

ARCD VS ADPKD IN END-STAGE RENAL DISEASE

	ARCD	ADPKD
Kidney size	Small, usually <300 g	Large, usually >800 g
Cysts	Small	Large
Extrarenal features	None	Liver cysts, aneurysms, etc.

Treatment

- **Bleeding:**
 1. **conservative** (bed rest, symptom control);
 2. **persistent bleeding or pain → nephrectomy or embolisation;**
 3. **haematuria with dialysis heparin → switch to peritoneal dialysis;**
 4. **retroperitoneal haemorrhage → nephrectomy if carcinoma cannot be excluded** (high risk of occult RCC).
- **Painful large cysts → percutaneous aspiration with cytology:** a temporising measure, since ARCD may regress after transplant.
- **Infected cyst:** percutaneous drainage → surgical drainage or nephrectomy if that fails.
- **Tumours:**
 - **>3 cm → surgical excision** (tumours <3 cm rarely metastasise, but small size is no guarantee);
 - **<3 cm → nephrectomy in fit patients, or annual CT with resection if enlarging;**
 - **laparoscopic bilateral radical nephrectomy** is proposed for ESRD with ACKD and suspicious tumours.

- **Screening** — its benefit is debated:
 - one approach: **after >3 years of haemodialysis**, ultrasound and CT, then **6-monthly ultrasound if clear**, or **6-monthly ultrasound plus CT if cysts or tumours <2 cm**;
 - restrict screening to **patients <55 years, on dialysis ≥3 years, in good health**;
 - or screen by **risk factors: prolonged dialysis, ARCD present, male sex**.
- **After transplantation:**
 - RCC can arise in native kidneys **3–8 years** later, especially in **older and male** recipients;
 - **immunosuppression adds risk**;
 - **native kidneys = 4.5% of malignancies in transplant recipients; >70% of RCCs in transplant patients arise in native kidneys rather than the graft**.

BEHAVIOUR OF RCC — HAEMODIALYSIS VS TRANSPLANT POPULATIONS

Haemodialysis	Transplantation
RCC rarely metastasises or kills; its clinical significance is questioned	Aggressive → evaluate thoroughly before and after transplant

KEY POINTS: ACQUIRED RENAL CYSTIC DISEASE

- **Associated with chronic renal failure**, mostly long-term haemodialysis and peritoneal dialysis.
- **Hyperplastic cysts and frequent adenomas; either can progress to RCC**.
- **Renal malignancy in dialysis patients 5–50× the general population**; RCC mostly arises **before 10 years of dialysis**.
- **Cysts usually regress with transplantation, but RCC risk persists at a lower level**.

Concept — *ARCD is a disease of uraemia, not of dialysis. It grows with time in the uraemic state, and malignancy tracks that time. Transplantation may remove the stimulus but not the cells already transformed, and immunosuppression then turns an indolent dialysis cancer into an aggressive transplant cancer.*

CALYCEAL DIVERTICULUM (PYELOGENIC CYST)

- **Outpouching of the collecting system into the corticomedullary region, usually from a calyceal fornix**. It is a **transitional epithelium-lined, smooth, spherical cavity** joined by a **thin neck**, typically at the **upper or lower pole**.
 - "Pyelogenic cyst" especially if it arises from the pelvis.
 - Size: millimetres to centimetres.
- **Mostly congenital**; similar lesions follow **blunt trauma** or **infundibular obstruction**.
- **Usually asymptomatic**; may contain **one large stone, many tiny stones or milk of calcium**. Stone passage through the neck → **pain and haematuria**.
- **Seen on CT, MRI and ultrasound, but distinguishing it from an obstructed calyx or a renal cyst needs contrast: delayed contrast layering confirms it (Fig. 50.29). See Chapter 49.**

PARAPELVIC AND RENAL SINUS CYSTS

- **Many synonyms**: parapelvic, renal sinus, peripelvic, pelvic, parapelvic lymphatic and hilar cysts, and parapelvic lymphangiectasia.

PARAPELVIC (PERIPELVIC) CYST VS RENAL SINUS CYST

	Parapelvic / peripelvic cyst	Renal sinus cyst
Origin	Simple parenchymal cyst impinging on the pelvis	Sinus structures (lymphatics, arteries, fat) — no parenchymal origin
Correct use of the term	Describes location of a parenchymal cyst	Describes all other hilar cysts
Mechanism	As a simple cyst	Mainly dilated lymphatics ; also fluid replacing sinus fat as the kidney atrophies from vascular disease

- **Benign; 15–20% at autopsy. Almost always asymptomatic**; rare collecting-system obstruction.
- **Ultrasound**: multiple echo-free sinus areas that **mimic hydronephrosis. Hydronephrosis is certain only when dilated calyces are seen to communicate with a dilated pelvis.**
- **CT**: water attenuation; cysts **displace calyces peripherally**. On unenhanced scans multiple sinus cysts mimic a dilated pelvis; **delayed contrast in an undilated collecting system settles it** (Fig. 50.30).
- **Differential diagnosis**: renal sinus lipomatosis (fat density), lymphoma, haemorrhage, urinoma.
- **Management**: conservative.